UGC NET June 2024 - Computer Science & Applications (Subject Code 87)

Official PYQ Paper - University Grants Commission National Eligibility Test (JRF)

Q1. Consider the following scheduling algorithms: (I) First-Come, First-Served (FCFS), (II) Shortest Job First (non-preemptive), (III) Round Robin (RR), (IV) Shortest Remaining Time First (SRTF). Which of the above scheduling algorithms are non-preemptive?

(1) (I) and (II) only
(2) (I), (II) and (IV) only
(3) (II) and (III) only
(4) (I) and (IV) only

Q2. In relational database theory, which of the following is NOT one of Codd's fundamental relational algebra operators?

(1) Selection
(2) Projection
(3) Natural Join
(4) Cartesian Product

Q3. The language L = {a^n b^n c^n | n >= 1} is:

(1) Regular
(2) Context-Free but not regular
(3) Context-Sensitive but not context-free
(4) Not recursively enumerable

Q4. What is the worst-case time complexity of searching an element in a balanced AVL search tree with n keys?

(1) O(1)
(2) O(log n)
(3) O(n)
(4) O(n log n)

Q5. An IPv4 datagram has arrived with the first 8 bits having the hexadecimal value 45. What is the header length in bytes?

(1) 5 bytes
(2) 20 bytes
(3) 45 bytes
(4) 40 bytes